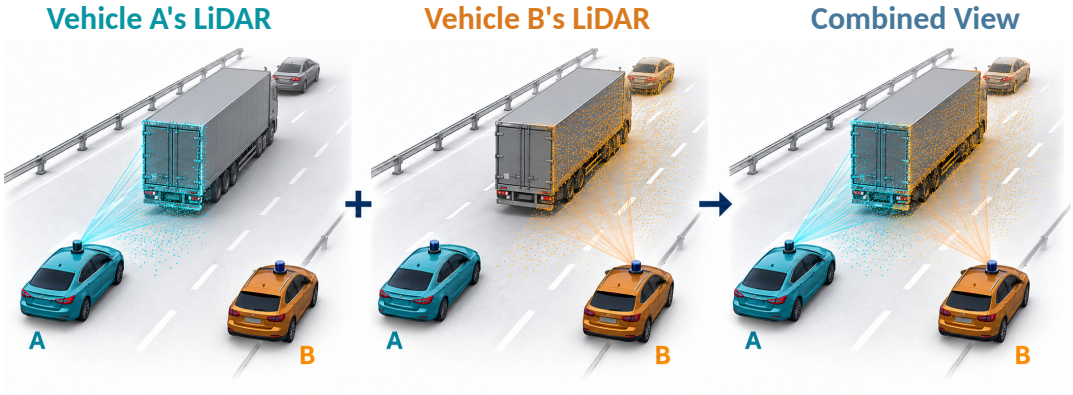
MO809: Course Project Proposal

Cooperative Perception with DiscoNet under Perfect Communication

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Local vs Cooperative



Truck Blocks A's View

B Sees the Hidden Car

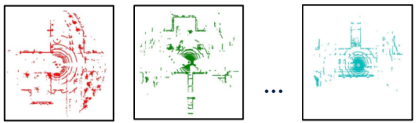
Shared LiDAR Observations

AI-generated illustration.

LiDAR View Aggregation

LiDAR Points in Bird's-Eye View (BEV)

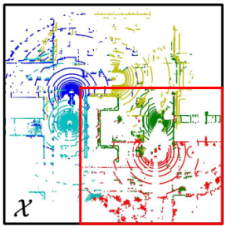
1. Local Views



Agent 1 Agent 2 Agent M

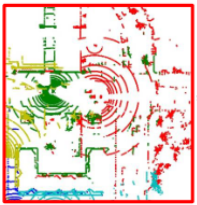
Each vehicle's own points.

2. Align and Combine



Align using poses into one frame.

3. Ego-Centered Crop

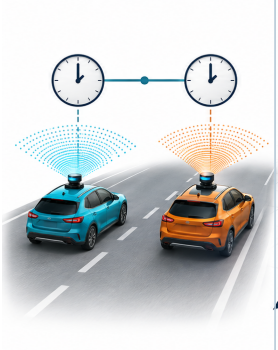


Agent 1's frame, cropped to its region.

Source: Li et al. (2021), Fig. 2.

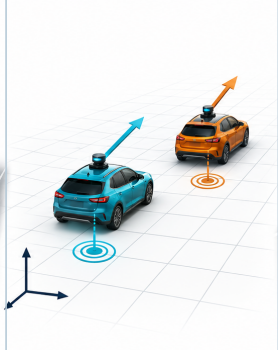
Perfect Communication

Synchronized



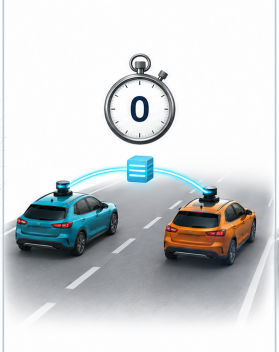
Same observation time

Accurate Poses



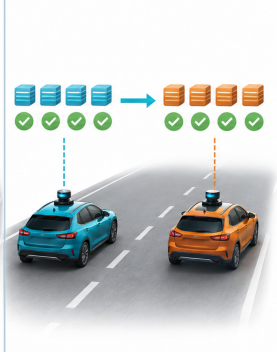
Position and heading

Zero Delay



0 ms latency

No Packet Loss

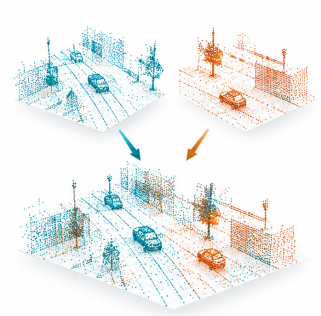


0% packet loss

AI-generated illustration.

Fusion Strategies

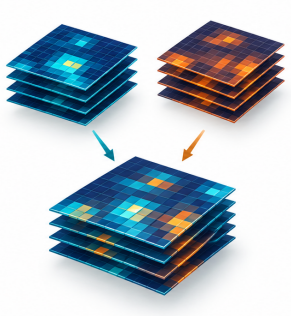
Early Fusion



Raw LiDAR Points

Fuse points, then detect

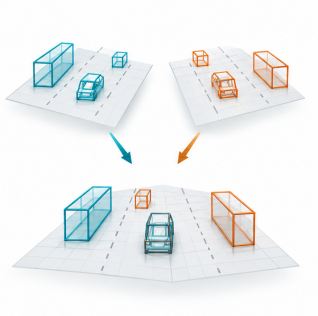
Intermediate Fusion



Learned Features

Encode → Fuse → Detect

Late Fusion

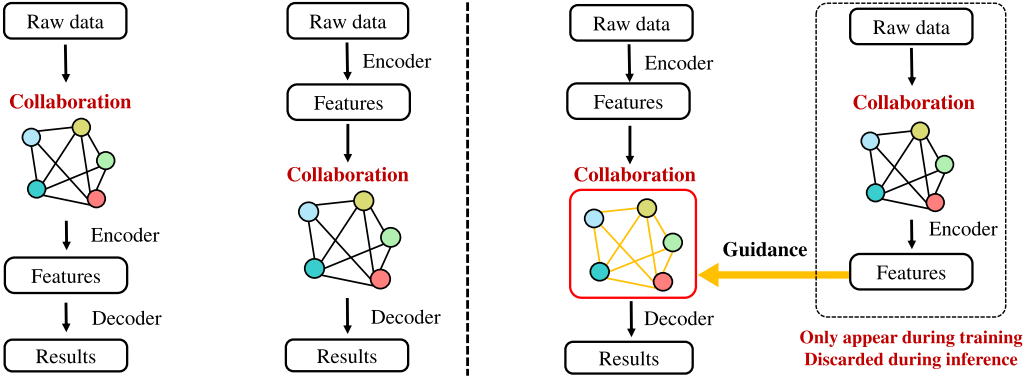


Boxes, Classes, Scores

Detect locally, then fuse

AI-generated illustration.

Collaboration Schemes

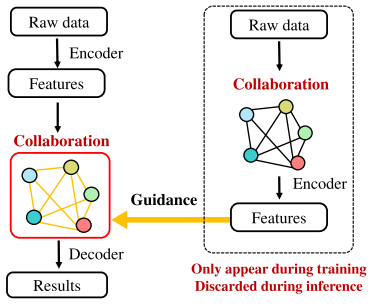


(a) Early (b) Intermediate (c) Ours

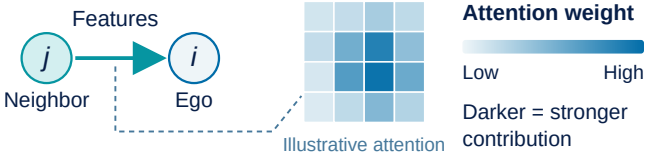
Source: Li et al. (2021), Fig. 1.

How DiscoNet Works

DiscoNet: Distilled Collaboration Network



Feature Exchange and Spatial Attention



Share learned BEV features.

Compute attention matrices at the receiver.

Knowledge Distillation (KD)

The teacher guides student features during training.

Source: Li et al. (2021), Fig. 1(c).

Experiments and Tools

1. Detection Quality

- AP @ 0.5
- AP @ 0.7

2. Datasets

- V2X-Sim
- OPV2V

3. Simulators

CARLA



Urban driving simulation

AP: Average precision
IoU: Intersection over union

Thanks!

Questions?